NLP CS60075 · ASSIGNMENT II · PROJECT 9 · GROUP 27

Parameter-Efficient Fine-Tuning on IMDB

From Hand-Engineered Lexical Features to Low-Rank Adaptation (LoRA) on Pretrained Transformers

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Low-Rank Updates Rival Full Fine-Tuning

Full BERT Fine-Tuning

- **Parameter Cost:** Updates all 109.48M backbone weights.
- **Storage Footprint:** Requires 440 MB checkpoint per task.
- **Training Overhead:** Needs gradient accumulation on 16GB T4.
- **Context Limit:** Truncates reviews exceeding 512 subwords.

LoRA + Classical Fusion

- **Low-Rank Update:** Trains only 0.14% to 1.07% weights.
- **Modular Storage:** Saves lightweight megabyte adapters.
- **Full-Text Signal:** TF-IDF captures cues past 512 tokens.
- **Core Question:** Do lexical counts improve adapted BERT?

Holdout Split Prevents Stacking Leakage

STANFORDNLP/IMDB
50,000 Reviews

Balanced polarity with HTML tags cleaned.

→

CORE TRAIN (90%)
22,500 Reviews

Fits Task 1 RF and Task 2 LoRA models.

→

HOLDOUT SPLIT (10%)
2,500 Reviews

Selects LoRA rank and fits Task 3 stack.

→

OFFICIAL EVALUATION
25,000 Test

Touched once for final benchmark metrics.

BERT 512-Subword Truncation Impact: 13.3% Train & 12.3% Test Reviews Truncated

Median Length: 222 Train / 219 Test Subwords (Seed = 42)

Lexical N-Grams Reach 84.77% Accuracy

SPEC-EXACT TEST ACCURACY

84.77%

Word 1–2 grams (20k cap) with 300-tree Random Forest (depth 30, min leaf 5).

Baseline Variant	Train Split	Holdout Acc	Test Acc
TF-IDF + RF (Full Spec)	**25,000**	**n/a**	**0.8477**
TF-IDF + RF (Core Split)	22,500	0.8528	0.8439
Negative Class Precision / Recall	22,500	0.8651 (P)	0.8150 (R)
Positive Class Precision / Recall	22,500	0.8251 (P)	0.8729 (R)

Key Observation: Shallow depth-30 trees produce cautious probabilities clustered near 0.5 and struggle with long-scope negation.

Low-Rank Matrices Compress BERT Updates

$$h = W_0 x + \frac{\alpha}{r}\Delta W x = W_0 x + \frac{\alpha}{r} B A x \qquad (B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times d})$$

Pretrained weights **W0** stay frozen; Query and Value projections across all 12 layers learn rank-**r** updates (**2dr** params).

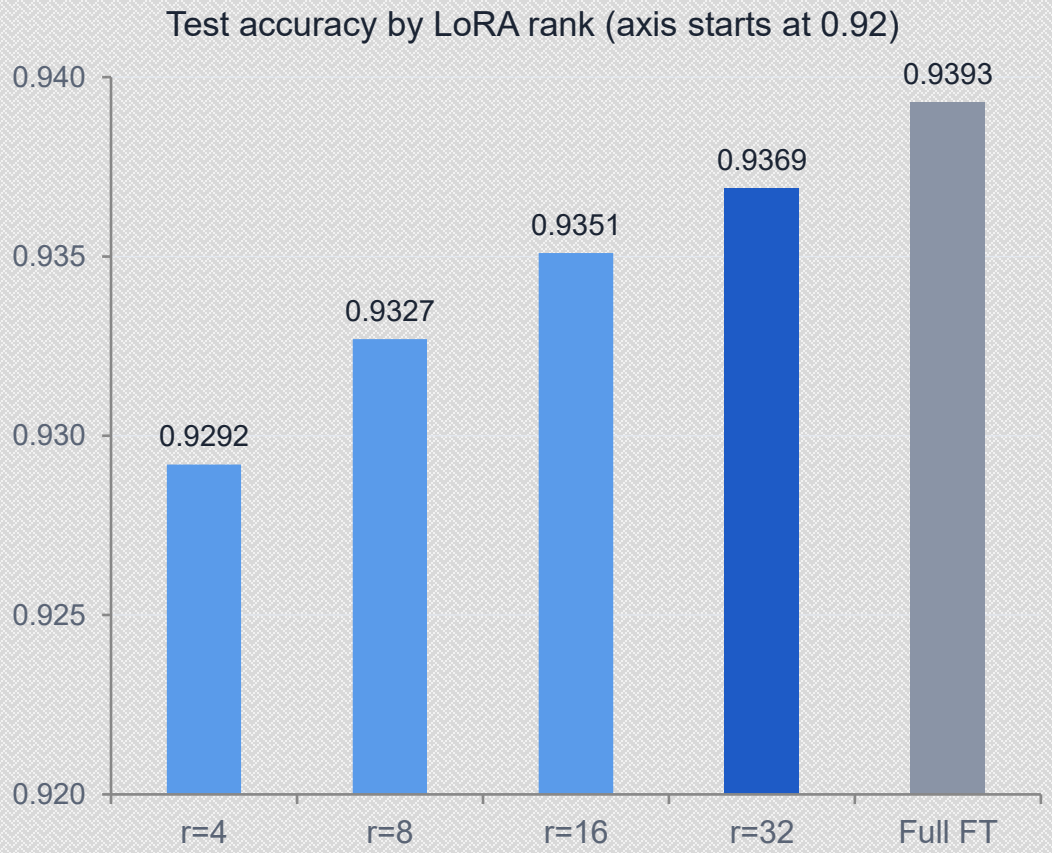
```
# Task 2b: PEFT LoRA Configuration on bert-base-uncased ( IMDB_Assignment.ipynb )


base = AutoModelForSequenceClassification.from_pretrained("bert-base-uncased", num_labels=2)
cfg = LoraConfig(
    task_type=TaskType.SEQ_CLS, r=r, lora_alpha=2 * r,
    lora_dropout=0.1, target_modules=["query", "value"]
)
model = get_peft_model(base, cfg) # AdamW lr=3e-4, batch_size=16, epochs=3, fp16 AMP
```

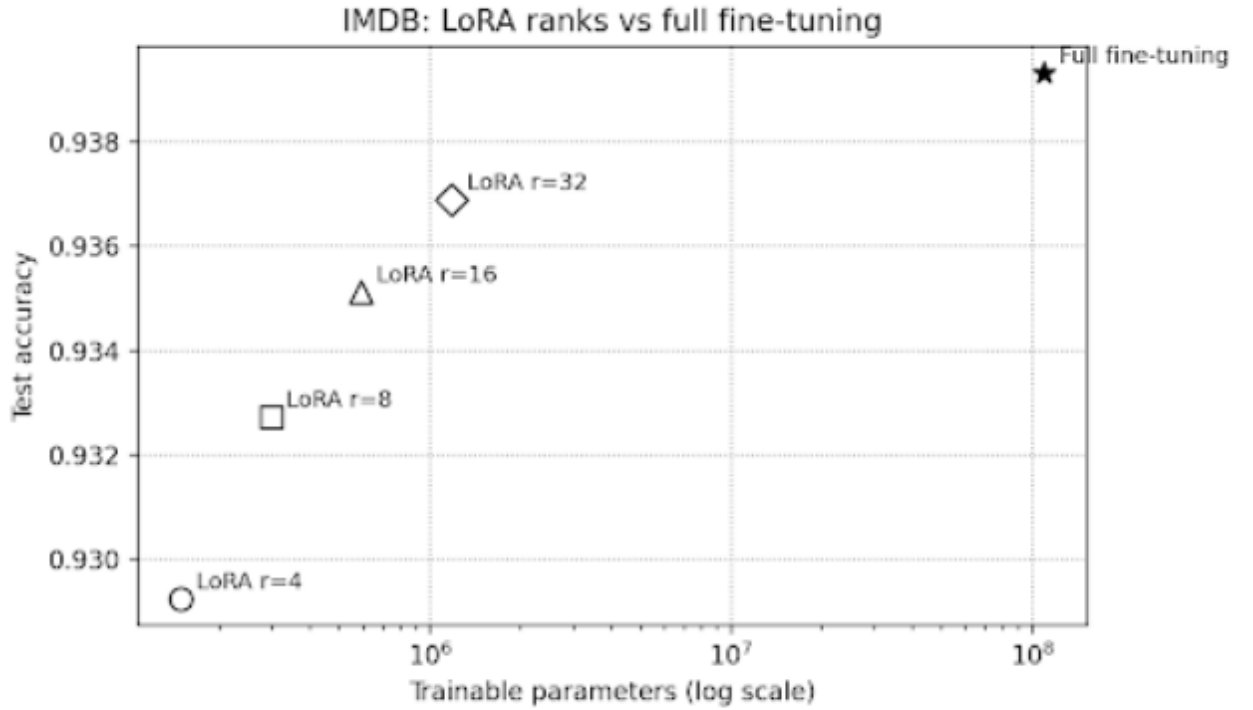
LoRA at r = 32 lands within 0.24 points of full fine-tuning with 1% of the weights

Model	Trainable	Test acc	Min
LoRA r=4	148,994	0.9292	26.2
LoRA r=8	296,450	0.9327	26.3
LoRA r=16	591,362	0.9351	26.3
LoRA r=32	1,181,186	**0.9369**	26.3
Full fine-tuning	109,483,778	**0.9393**	31.5

Gain per rank doubling: +0.35, +0.24, +0.18 points



Diminishing Returns Above 1% Parameters



Empirical Scaling Laws

- **93x Compression:** Rank 32 lands within 0.24 pts of full FT.
- **735x Compression:** Rank 4 closes 89% of baseline gap.
- **Sublinear Gains:** Doubling rank adds +0.35, +0.24, +0.18 pts.
- **Low Intrinsic Rank:** Confirms Hu et al. (2022) hypothesis.
- **Holdout Variance:** 0.51-pt standard error explains r=8 dip.

Log-Odds Stacking Beats Early Fusion

LATE FUSION (MAIN WINNER)
Log-Odds Stacking

Base Models
Frozen RF + LoRA r=32

→

Logit Transform
log(p / (1 - p))

→

LR Meta-Model
93.94% Test (3 Params)

EARLY FUSION (STRETCH)
Joint End-to-End Net

Dual Inputs
20k TF-IDF + [CLS] (768d)

→

Dense Projection
5.12M Weights → 256d

→

Lexical Overfit
93.49% Test (6.3M Params)

Log-Odds Meta-Model Matches Full BERT FT



Paired Bootstrap Gain (Task 3 − Task 2): +0.26 pts (95% CI [+0.13, +0.40])

Exact McNemar Test: p = 0.000184 (175 Fixed vs 111 Broken)

RF Breaks Ties When BERT Is Uncertain

MODEL DISAGREEMENTS (3,678)

83.36%

Outperforms LoRA 81.59% solo accuracy.

TRUNCATED REVIEWS (3,086)

+0.36 pts

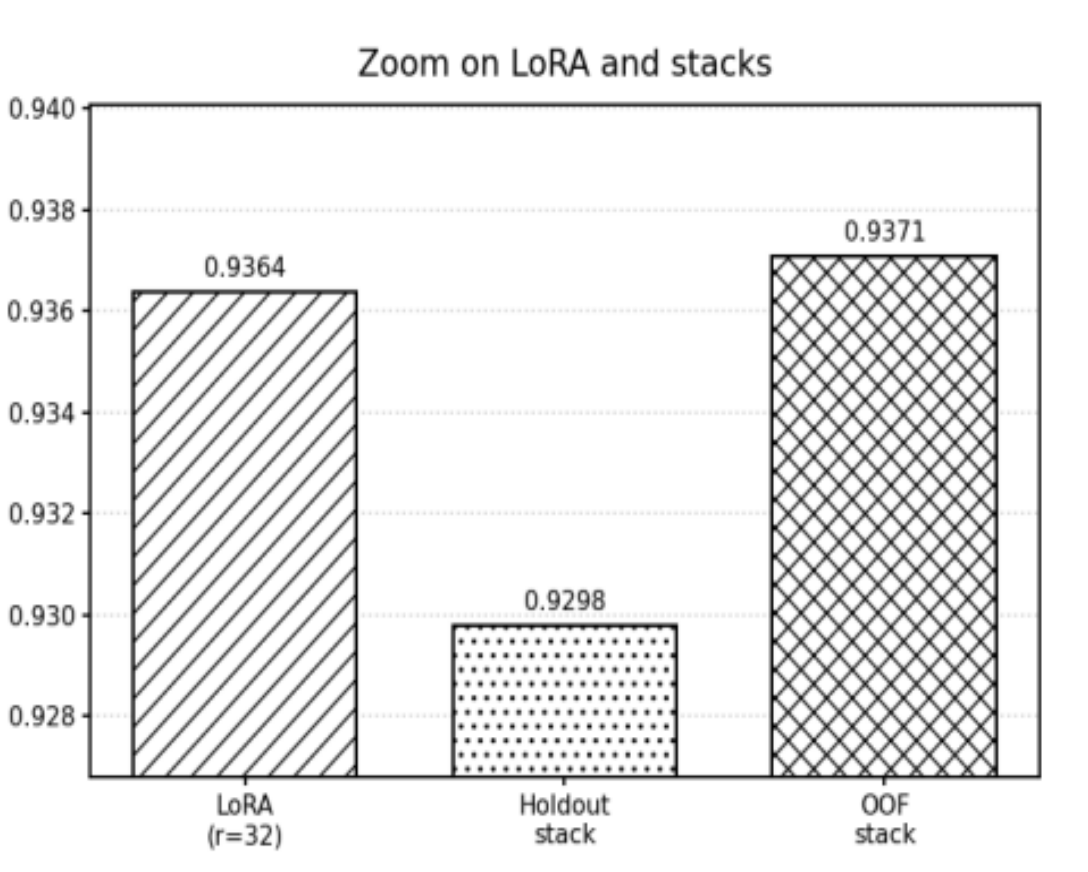
Full-text TF-IDF recovers cut endings.

LEARNED LOG-ODDS WEIGHTS

1.80 / 0.50

Rescales cautious near-0.5 RF outputs.

OOF Stacking Matches Solo LoRA on Raw Probs



Test accuracy of the LoRA r=32, the holdout stack and the OOF stack in the second notebook.

Observations :

- **Marginal Gain:** The OOF stack's 0.07-point accuracy increase over LoRA alone is within statistical noise.
- **Signal Waste:** Feeding raw probabilities to the meta-model is ineffective because LoRA outputs sit near 0 or 1.
- **High Cost:** The method requires six LoRA trainings, costing roughly six times the GPU time of the holdout approach.
- **Takeaway:** Using an OOF meta-model on raw probabilities yields results level with using LoRA alone.

Qualitative check: the stack only overrides LoRA when LoRA is unsure

LoRA confident error

gold negative · p_pos 0.999

“This movie was pure genius…”

Reads positive: likely label noise

Correct but unsure

gold positive · p_pos 0.500

“…an easy decision not to see this film…”

Negative opening that turns positive

Stack fixed LoRA

p_rf 0.392 · p_lora 0.757 → p_meta 0.438

“I wasn't sure who this film was aimed at…”

RF leaned right while LoRA was unsure

Stack broke LoRA

p_rf 0.578 · p_lora 0.277 → p_meta 0.514

“An obvious vanity press for Julie…”

Sarcastic summary fools word counts

Key Takeaways & Open Questions

97.5%

Full FT Gap Closed at 1% Params

93.94%

Log-Odds Meta-Stack Test Accuracy

0.00018

McNemar p-Value vs Solo LoRA

Late log-odds stacking preserves parameter efficiency and avoids the 5.1M-parameter lexical overfitting of joint early-fusion networks.